
Uncovering Temporal Dynamics in SEP EEG Data

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1 Introduction

In the summer 2024 from the 1st of July to the 31st of September I conducted an internship at the Max Planck Institute (MPI) for Human Cognitive and Brain Sciences in Leipzig. During that time I was closely working with my supervisors Prof. Dr. Thomas Knösche and Sasha Smolarchik Brenner Socas, who is doing his Phd at the MPI.

The project I was assigned to focused on analyzing the temporal dynamics of electroencephalography (EEG) recordings during repeated sensory stimulation. From neuroscience, it is well established that the human brain exhibits characteristic responses to such stimulation, which manifest on average as somatosensory evoked potentials (SEPs) [1]. The fact that SEPs reliably appear after averaging suggests that individual EEG signals reflect noisy realizations of an underlying generative process governing the neural response to somatosensory stimulation [2].

My task was to investigate whether Machine Learning techniques could be used not only to identify dominant temporal modes in the data, but also to uncover aspects of this generative process itself, thereby providing deeper insight into the underlying temporal dynamics of the SEP EEG signals. This required familiarization with the dataset and an exploration of suitable methodological approaches. To this end, we conducted two complementary analyses. First, we applied Principal Component Analysis (PCA) to identify dominant modes in the EEG signals and assessed to what extent these components capture meaningful temporal structure. Second, we trained a transformer-based model for signal forecasting on synthetically generated data designed to mimic key characteristics of our EEG recordings. This enabled us to assess whether the model can recover the parameters of the underlying distribution from which the synthetic data were generated. Together, these analyses provide an initial perspective on the challenges involved in uncovering the temporal dynamics of SEP-related EEG data using Machine Learning.

I also dedicated part of my internship to theoretical research, studying the fundamentals of dynamical systems and the potential of data-driven analyses of these systems. I studied the work of Geneva and Zabaras [3] in depth and discussed the potential of Koopman embeddings for improving transformer models applied to dynamical systems. They serve as informative embeddings for transformers to facilitate the prediction of the evolution of dynamical systems. Additionally, I familiarized myself with the fundamentals of Dynamic Mode Decomposition, a classical dimensionality reduction method for time-series data, by studying Chapter 7.2 of the book by Kutz et al. [4].

All code used for this project, including the model training, evaluation, and visualizations was implemented in Python. The full implementation is publicly available at https://github.com/allshaks/eeg_llm.

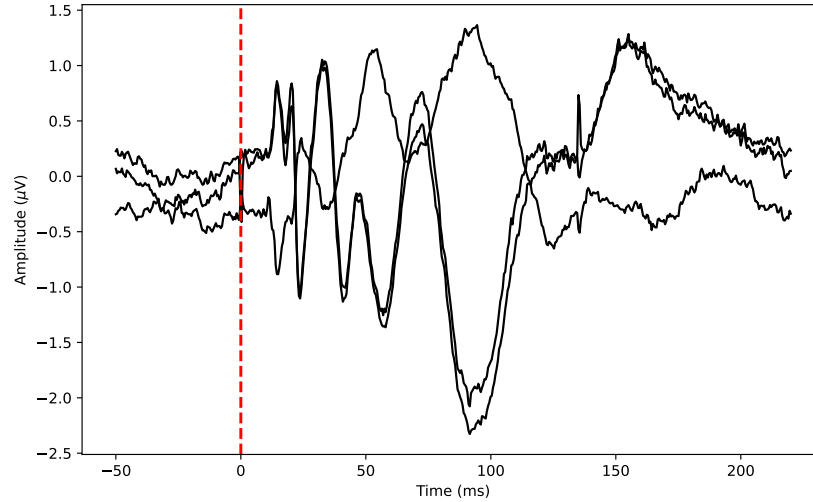


Figure 1: Somatosensory evoked potentials of different channels are shown in black. The dotted horizontal line indicates the onset of the stimulus.

2 Theoretical Background

2.1 Somatosensory Brain Responses

Electroencephalography (EEG) is commonly used to record the brain’s response to sensory stimuli, such as an electrical pulse applied to a peripheral nerve in the forearm or leg. EEG is a non-invasive technique that measures the brain’s electrical activity through electrodes placed on the scalp [5]. In EEG terminology, we often refer to channels rather than electrodes. Each channel records the voltage difference between two electrodes or between an electrode and a common reference, e.g., the average of all electrodes. The arrangement of these channels across the scalp is referred to as the montage.

When a subject is repeatedly stimulated, experiments have shown that so-called SEPs emerge when averaging across many trials [1]. Each trial, also called an epoch, corresponds to a single presentation of the stimulus. Individual trials are typically extremely noisy and dominated by background EEG activity unrelated to the stimulus. However, when averaging a sufficiently large number of trials, this noise largely cancels out and characteristic SEP waveforms become visible. The waveform of SEPs differs across channels, as different cortical regions contribute differently to the sensory response, as shown in Figure 1.

To investigate whether different channels share similar temporal dynamics, Principal Component Analysis (PCA) provides a suitable framework. It identifies the most dominant temporal modes present across all channels.

2.2 Identifying Dominant Temporal Patterns

PCA is a widely used linear dimensionality reduction method, originally introduced by Pearson [6] and later independently formalized and named by Hotelling [7]. PCA is commonly employed for feature extraction and visualization of high-dimensional data, but can also be used to identify dominant modes in time-series signals. This is particularly relevant for our setting: SEPs exhibit characteristic temporal waveforms across channels, and it is of interest to determine whether these channels share a common dominant pattern.

As described in Chapter 1.5 of Kutz et al. [4], PCA identifies orthogonal directions in the data that capture the maximal amount of variance. These directions, called principal components, are obtained by solving the eigenvalue problem of the mean-centered data matrix. The components are ordered

according to the proportion of variance they explain. Projecting the data onto the leading principal components yields a lower-dimensional representation that preserves the most informative structure in the original data.

While PCA provides a good starting point for identifying global temporal structure in the SEPs, it is limited to linear relationships and is typically applied to matrix-shaped data. SEP EEG recordings, however, consist of epochs, each containing multiple time points across many channels, resulting in data that cannot directly be expressed as a matrix. For these reasons, we turn to transformer models, a powerful class of neural-network-based Machine Learning methods capable of modeling nonlinear relationships and flexible with respect to their input structure. This enables us to investigate whether more expressive models can recover aspects of the underlying generative process and temporal dynamics of the EEG signals.

2.3 Time Series Forecasting

Transformers are a class of deep learning architectures first introduced by Vaswani et al. [8] and initially developed for natural language processing tasks. They rely on a mechanism known as attention, which enables the model to process an entire input sequence in parallel and to learn weighted dependencies between elements at different positions within that sequence. Through these learned attention weights, the model can selectively focus on relevant parts of the input when generating predictions.

Transformers are particularly well suited for time series modeling because attention allows them to capture both short- and long-range temporal dependencies without relying on sequential processing, as required by other recurrent architectures [9]. They are also flexible with respect to input structure, accommodating multivariate time series as well as sequences of variable length. In contrast to classical point-valued forecasting, transformer-based time series models perform probabilistic forecasting: instead of predicting a single value, they output the parameters of a probability distribution from which predictions can be sampled.

In the context of our EEG data, these properties can be exploited by providing the model with a portion of each signal as context and letting it predict the distributions of the remaining time points. If the predicted distributions match those underlying the data, this would indicate that the transformer has successfully captured key aspects of the generative process.

3 Material and Methods

3.1 Dataset

As a dataset, we used SEP EEG recordings obtained from Graichen et al. [10]. In this experiment, the median nerve in the right forearm of 11 healthy male volunteers was stimulated at a rate of 3.7 Hz. Participants were between 26 and 40 years old (mean 30.7 ± 4.3 years). EEGs were recorded using an ANT Neuro BV Waveguard 256-channel EEG cap at a sampling frequency of 2,048 Hz. The dataset was cleaned and artifact-checked by the original authors.

For our analyses, we focused on one participant (Subject 6). This participant’s recording contains 6,038 epochs, each comprising 554 time points equaling 270 ms per epoch. Thus, the total recording time corresponds to approximately 27.2 minutes of stimulation and results in about 3.5 GB of data.

3.2 Data Preprocessing

As the original authors had already performed artifact removal, we did not implement further preprocessing. We simply loaded the data and segmented it into epochs corresponding to what we described above. The resulting dataset can be represented as

$$\mathcal{D} = \left(\mathbf{E}^{(1)}, \mathbf{E}^{(2)}, \dots, \mathbf{E}^{(N)} \right),$$

where N is the number of epochs and each epoch $\mathbf{E}^{(i)} \in \mathbb{R}^{C \times T}$ contains C channels and T time points. In our data, $N = 6038$, $C = 256$, and $T = 554$.

3.3 Principal Component Analysis

To investigate the dominant temporal dynamics in the SEP data, we applied PCA. Since our goal was to extract global temporal structure across channels, and PCA requires matrix-shaped input whereas SEP data does not (see Section 2.2), we averaged the EEG recordings. This resulted in an average data matrix

$$X \in \mathbb{R}^{C \times T},$$

where the i -th row corresponds to the mean time series of the i -th channel.

PCA was then performed on this matrix to extract the directions of maximal variance in the averaged data. In this setting, the resulting components correspond to dominant temporal modes that are expressed to varying degrees across different channels.

We first examined the explained variance ratio of each loading, taking k loadings that together account for 95% of the total variance into account. Subsequently, we assessed the similarity between the first principal component and the average waveform of each channel by computing the Pearson correlation coefficient (PCC). The spatial distribution of these correlations was visualized using a topographic map. Finally, to evaluate whether the dominant temporal mode identified from the averaged data is also present at the single-trial level, we computed the PCC between the first temporal mode and each channel’s time series in every epoch. We then compared the mean absolute correlations from single trials to those derived from the averaged data and visualized representative results as topographic maps.

While PCA allowed us to identify dominant modes of the SEPs across channels, we also aimed to investigate the temporal dynamics at the single-trial level. To this end, we employed time series transformers for per-trial signal forecasting in order to assess whether they could capture aspects of the underlying generative process.

3.4 Time Series Transformers

Since the generative process underlying EEG recordings is not known, we evaluated the performance of transformer models on a synthetically generated dataset. This served as a preliminary step toward SEP EEG signal prediction. At the time of the project, applying a Transformer directly to the full SEP dataset would have been difficult due to computational constraints. The synthetic data was designed to mimic key characteristics of EEG signals, such as oscillatory structure, additive noise, and temporal shifts, while allowing full control over the underlying distribution.

SEP EEG signals can be regarded as random realizations from an underlying generative process, with additional noise introduced by measurement and physiological variability. To reflect this assumption, we generated synthetic data consisting of Gaussian-shaped pulses that were randomly time-shifted and corrupted by additive noise.

Formally, a synthetic signal for channel i is defined as

$$p_i[n] = g(n - \Delta; \mu_i, \sigma_i) + \alpha \epsilon_i[n], \quad \epsilon_i[n] \sim \mathcal{N}(0, 1), \quad \Delta \sim \mathcal{N}(0, 1),$$

where n indexes discrete time points, μ_i and σ_i denote the mean and standard deviation of the Gaussian pulse $g(\cdot; \mu_i, \sigma_i)$, Δ is a random time shift sampled once per epoch, and α controls the amplitude of the additive noise $\epsilon_i[n]$. A synthetic epoch is then constructed as

$$E^{(j)} = \begin{pmatrix} p_1[n] \\ p_2[n] \\ \vdots \\ p_C[n] \end{pmatrix},$$

where each $E^{(j)} \in \mathbb{R}^{C \times T}$ contains all C channels over T time points. The complete synthetic dataset is given by

$$\mathcal{D} = \left(E^{(1)}, E^{(2)}, \dots, E^{(N)} \right).$$

We generated $N = 1000$ i.i.d. synthetic epochs, each consisting of $C = 2$ channels and $T = 100$ time points. For the two channels, we specified different mean and standard deviation parameters for the underlying Gaussian pulses:

$$\begin{aligned} \mu_1 &= 20, & \sigma_1 &= 6, \\ \mu_2 &= 40, & \sigma_2 &= 6. \end{aligned}$$

The noise weight was fixed to $\alpha = 0.5$ for all channels.

The dataset was randomly split into an 80% training set and a 20% test set. The model used for prediction was the *Time Series Transformer* from the Hugging Face `transformers` library. To configure the model, we initialized it using the `TimeSeriesTransformerConfig` class, where all relevant hyperparameters for training are specified. Each signal was divided into a context window and a prediction segment, where the Transformer received the first 50% of the sequence as input and was tasked to predict the remaining 50 time points. The model was trained using a batch size of 16, with two encoder and two decoder blocks. The number of attention heads was set to eight, the dimension of the model to 64, and training was performed for 20 epochs with a learning rate of 10^{-3} . The Transformer outputs the parameters of a probability distribution for each predicted time step. In our case, we selected a Gaussian distribution, meaning the model produces a mean and a standard deviation as outputs. During training, the negative log-likelihood is computed to measure how probable the observed data point is under the predicted distribution. This loss is then backpropagated through the network to update the model parameters accordingly.

At inference time, new samples can be generated by drawing from the predicted Gaussian distribution using the model’s outputted mean and standard deviation. To assess whether the Transformer was able to capture the underlying parameters governing the signal dynamics, we compared the true average of the synthetic signals with the average of the model’s predicted distribution parameters. If the predicted means and variances closely match the ground-truth parameters used to generate the data, this indicates that the Transformer has successfully inferred key aspects of the underlying generative process. Our analyses were restricted to evaluating whether the model recovered these global distributional parameters; we did not assess single-trial forecasting accuracy.

4 Results and Discussion

We found that the first seven principal components explain 95% of the total variance in the data. As illustrated in Figure 2, the variance is not evenly distributed among the components: the first principal component alone accounts for more than 50% of the variance, the second for approximately 25%, while all remaining components each explain less than 10% and often below 5%. This indicates the presence of one dominant temporal mode shared across most EEG channels.

To further characterize this dominant mode, shown in Figure 4, we computed the Pearson correlation between the time course of the first principal component and the average waveform of each channel. The resulting topographic map (Figure 3a) shows a clear dipolar pattern: electrodes over the left hemisphere exhibit strong positive correlations, whereas those over the right hemisphere are negatively correlated. The mean absolute correlation across all channels is 0.59 (rounded to two decimals). These results suggest that the first principal component captures meaningful large-scale dynamics present in the SEP data.

When repeating the correlation analysis on a per-trial basis, the clear dipolar organization disappears. As shown in Figures 3b, 3c, 3d, positively and negatively correlated channels appear more randomly distributed across the scalp, and many channels show only weak correlation. The mean correlation between the first principal component and individual trial channels decreases to 0.16, indicating that although traces of the dominant component remain in single trials, it explains substantially less of their variability compared to the averaged response. This discrepancy arises because single-trial SEPs have a low signal-to-noise ratio: the evoked response is small relative to the ongoing background EEG, which varies substantially across trials.

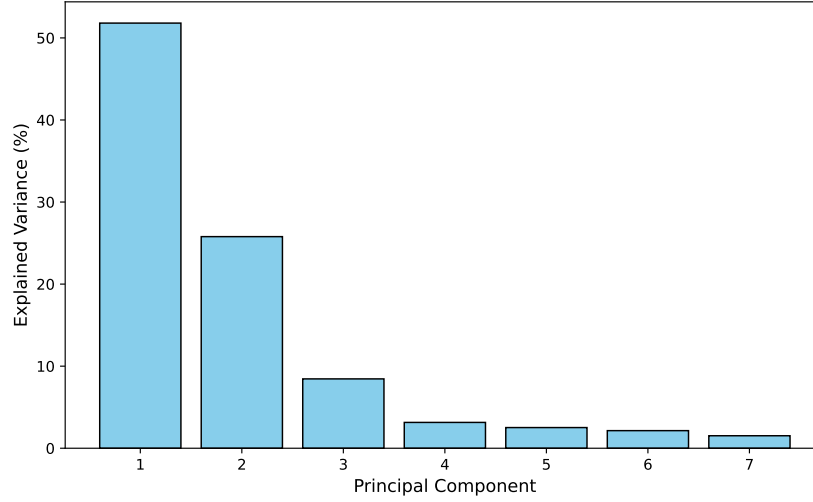


Figure 2: Explained variance of the first seven principal components accounting for 95% of the total variance.

Figures 5b, 5c, and 5d show examples of the synthetically generated signals. Due to the added noise and random temporal shifts, it is already challenging to visually determine the underlying distribution that generated the individual signals. However, when averaging across all samples, the expected distributions with means $\mu_1 = 20$ and $\mu_2 = 40$ and standard deviations $\sigma_1 = 6$ and $\sigma_2 = 6$ become clearly apparent, as shown in Figure 5a.

We next evaluated whether the Transformer could recover the global properties of the underlying generative process when trained on single-trial data in a signal forecasting task. Figure 6a illustrates the average predicted distribution parameters for each time point of the first channel. The learned means closely match the true average of the data, and the learned standard deviation for each time point is approximately 0.5, consistent with the noise level used during data generation. A comparable result is observed for the second channel, as shown in Figure 6b. This suggests that the Transformer successfully captured both the central tendency and variability of the signals, suggesting that it can effectively model the underlying generative process.

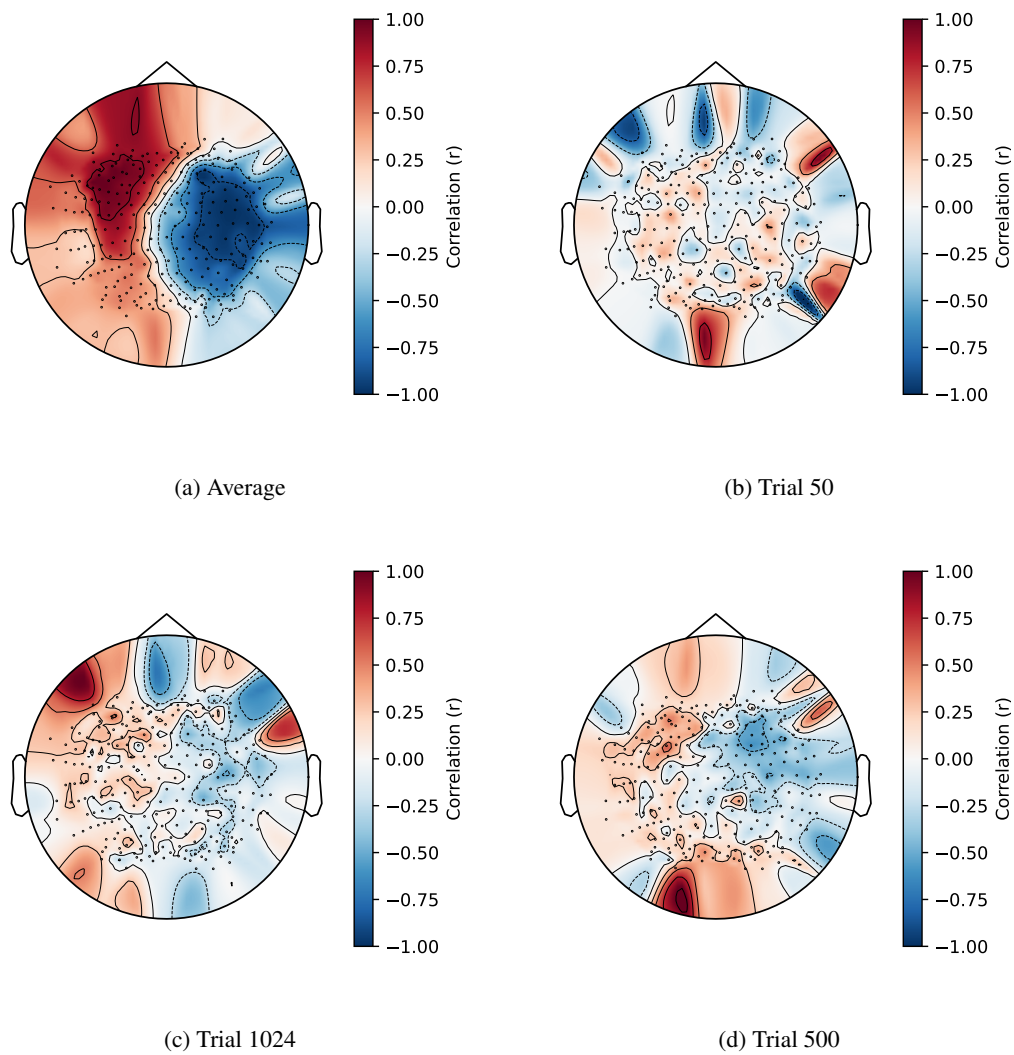


Figure 3: Topographic maps showing the Pearson correlation between the time course of the first principal component and the EEG signals. (a) displays correlations computed using the averaged SEP data, while (b), (c), and (d) show the corresponding correlations for three individual trials.

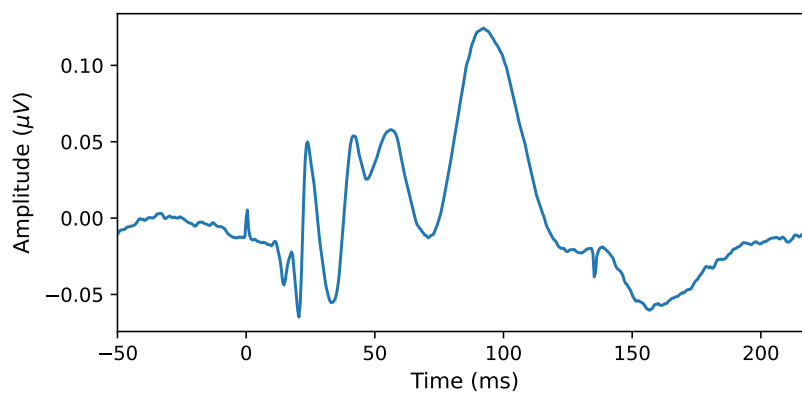


Figure 4: Wave form of the first principal component over time.

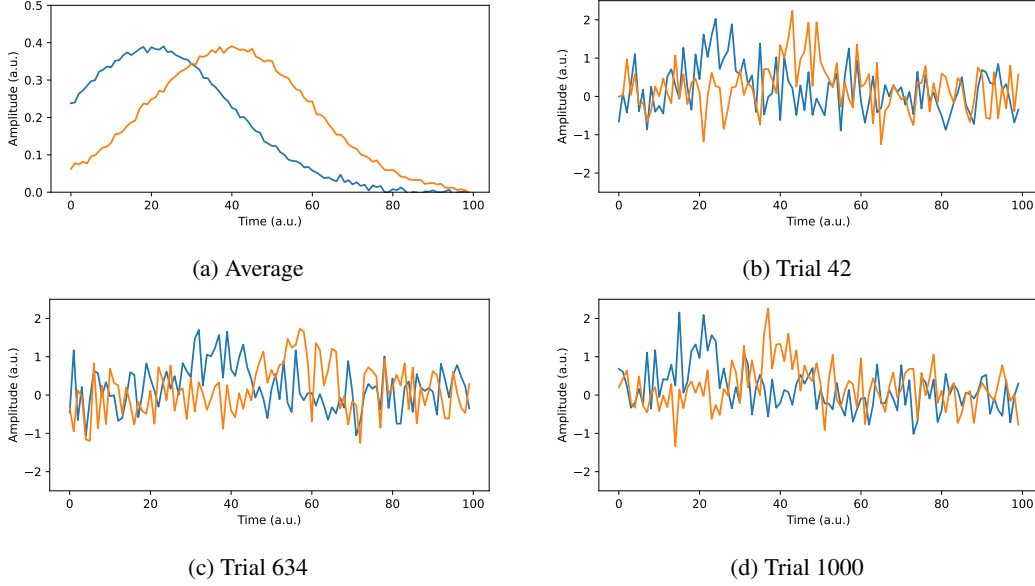


Figure 5: Example signals from the synthetic dataset. (a) shows the average across all generated trials, while (b)–(d) display three individual trials, illustrating the variability introduced by noise and temporal shifts. Channel 1 is shown in blue and Channel 2 in orange.

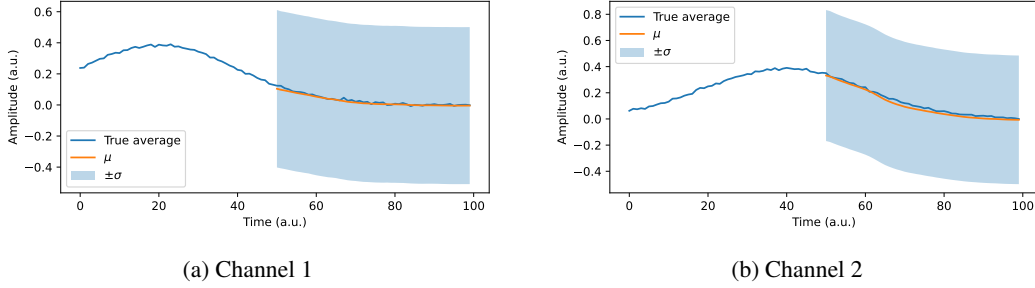


Figure 6: Learned distribution parameters of the Transformer model for both channels. For each forecasted time point, the plots show the average predicted mean and standard deviation across all predictions. The blue curve shows the true average of the data, the orange curve depicts the predicted means, and the light blue shaded region represents the predicted standard deviations.

5 Conclusion and Limitations

In this report, we demonstrated that PCA can be effectively applied to SEP data to extract the dominant temporal mode shared across channels. At the same time, our analyses revealed that this prominent mode of the averaged data shows only limited similarity to the behavior of individual trials, highlighting the difficulty of characterizing single-trial dynamics through linear methods alone.

Furthermore, we showed that a simple time series Transformer is capable of recovering the key distributional parameters underlying a synthetic generative process, using only individual samples and a forecasting objective. The synthetic data were designed to mimic the structure of SEP EEG recordings in spirit, and the results provide a first indication that Transformers may be suitable for modeling trial-wise SEP dynamics and for serving as generative models. However, this remains to be validated on real EEG data.

A natural next step would therefore be to train a Transformer directly on SEP EEG recordings and evaluate its forecasting performance. This would require more substantial computational resources,

which are available at the Max Planck Computing and Data Facility and which I had begun to familiarize myself with toward the end of the project.

In addition, several alternative dimensionality reduction techniques tailored to dynamical systems—such as Dynamic Mode Decomposition—could be explored to gain further insight into the dominant temporal structures of the data. Future work may also investigate the use of more informative embeddings for Transformers. Recent research suggests that embedding choices strongly influence model performance, and approaches such as Koopman-based embeddings could provide dynamic features that guide the model toward more meaningful representations [11], [3].

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